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Period 1

Creating Tetris

Description:

For my final project I would like to create a replication of the game tetris in processing. My code should be able to create blocks, rotate the blocks, move them down to their spot, and recognize when a row has been filled. Additionally, I need to include a point tracker as well as a Scores class to maintain a leaderboard. This leaderboard will likely have to be a local leaderboard stored on a .txt file.

I do not believe that I should need to use any outside libraries, since processing already has commands for graphics, keyboard input, a game loop, colors, etc.

How Does It Work:

Upon loading, a player will need to press the SPACE bar to begin the game. Then they will use the arrow keys to move the current piece left or right as it continuously moves down. Then once the piece touches another piece already on the board, it will become unmovable, and the next piece will appear. Additionally, if a player wants to switch between the current piece and the next piece they will be able to press the 'c' key to do so. To rotate the piece the player will press the 'r' key. The game will end when any set blocks touch the top.

Game
Board board Block currBlock Block nextBlock int score
game gameOver(): Boolean

Board
int[][] grid
placeBlock(): void detectCollision(): void detectLine(): void clearLine(): void

Leaderboard
ArrayList<ScoreEntry> scores
addScore(ScoreEntry): void sortScores(): void

Block
int xVal int yVal Boolean isMoving
moveDown(): void setLocation(int, int): void

ScoreEntry
String name int score
deleteScore(): void